LAB 2

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AI BTECH
B-11

12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

Program_12.c > main()

```c
#include <stdio.h>
int main()
{
    char ch;

    printf("Enter the character : ");
    scanf("%c", &ch);

    printf("ASCII code of %c is = %d", ch, ch);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ;
Enter the character : A
ASCII code of A is = 65
PS C:\Users\Lenovo\Desktop\C>
```

13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

Program_13.c > main()

```c
#include <stdio.h>
int main()
{
    int a;

    printf("Enter the value of a = ");
    scanf("%d", &a);

    if (a%2==0)
    {
        printf("Number is Even");
    }
    if (a%2!=0)
    {
        printf("Number is Odd");
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\"
Enter the value of a = 11
Number is Odd
PS C:\Users\Lenovo\Desktop\C>
```

14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

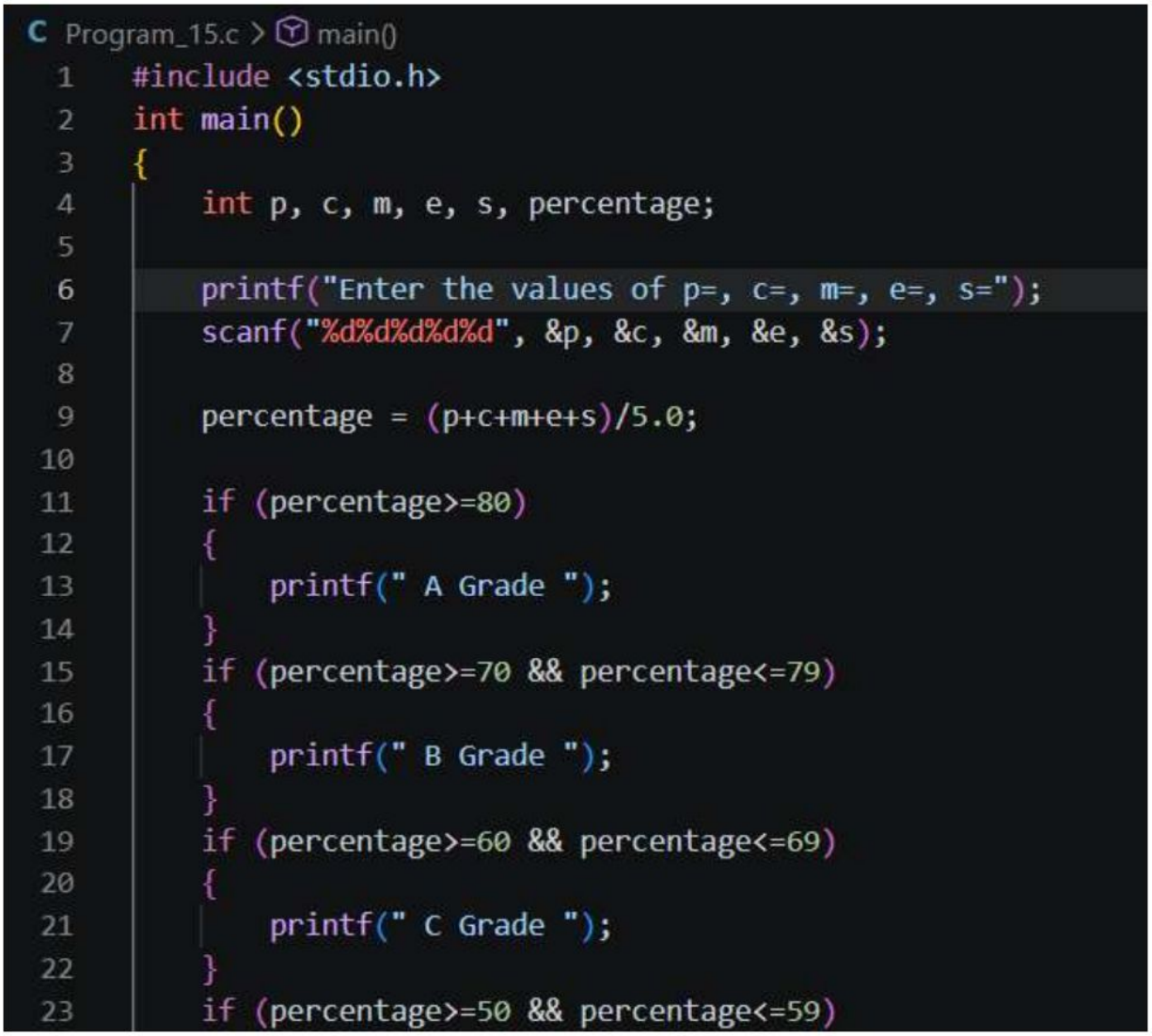
Program_14.c > main()

```c
#include <stdio.h>
int main(){
    char ch;
    printf("Enter the character : ");
    scanf("%c", &ch);
    if(ch>='A' && ch<='Z')
    {
        printf("Character is Capital letter");
    }
    else if(ch>='a' && ch<='z')
    {
        printf("Character is Small letter");
    }
    else if(ch>='0' && ch<='9')
    {
        printf("Character is Digit");
    }
    else
    {
        printf("Character is Special symbol");
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) {
Enter the character : S
Character is Capital letter
PS C:\Users\Lenovo\Desktop\C>
```

15. PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.



```c
#include <stdio.h>
int main()
{
    int p, c, m, e, s, percentage;

    printf("Enter the values of p=, c=, m=, e=, s=");
    scanf("%d%d%d%d%d", &p, &c, &m, &e, &s);

    percentage = (p+c+m+e+s)/5.0;

    if (percentage>=80)
    {
        printf(" A Grade ");
    }
    if (percentage>=70 && percentage<=79)
    {
        printf(" B Grade ");
    }
    if (percentage>=60 && percentage<=69)
    {
        printf(" C Grade ");
    }
    if (percentage>=50 && percentage<=59)
```

```
23      if (percentage>=50 && percentage<=59)
24      {
25          printf(" D Grade ");
26      }
27      if (percentage>=35 && percentage<=49)
28      {
29          printf(" E Grade ");
30      }
31      if (percentage<=35)
32      {
33          printf(" F Grade ");
34      }
35
36      return 0;
37  }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ;
Enter the values of p=, c=, m=, e=, s= 100 92 94 98 93
 A Grade
PS C:\Users\Lenovo\Desktop\C>
```

16. PROGRAM TO READ THREE NOS. AND PRINT MAX.

Program_16.c > main()

```c
#include <stdio.h>
int main()
{
    int a, b, c;

    printf("Enter the values of a= , b= ,c=");
    scanf("%d%d%d", &a, &b, &c);

    if(a>b)
    {
        if(a>c)
        {
            printf("a is max");
        }
        if(a<=c)
        {
            printf("c is max");
        }
```

```
    }
    if(a<=b)
    {
        if(b>c)
        {
            printf("b is max");
        }
        if(b<=c)
        {
            printf("c is max");
        }
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ;
Enter the values of a= , b= ,c= 16 77 52
b is max
PS C:\Users\Lenovo\Desktop\C>
```

17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR(&&).

Program_17.c > main()

```c
#include <stdio.h>
int main()
{
    int a, b, c;

    printf("Enter the value of a b c");
    scanf("%d%d%d", &a, &b, &c);
    if (a>b && a>c)
    {
        printf(" a is max ");
    }
    if (b>a && b>c)
    {
        printf(" b is max ");
    }
    if (c>a && c>b)
    {
        printf(" c is max ");
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?
Enter the value of a b c 23 4 86
 c is max
PS C:\Users\Lenovo\Desktop\C>
```

18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL OPERATOR(?:).

Program_18.c > main()

```c
#include <stdio.h>
int main()
{
    int a, b, c, max;

    printf("Enter the values of a ,b ,c ");
    scanf("%d%d%d", &a, &b, &c);

    max = (a>b) ? a : b;
    max = (max>c) ? max : c;

    printf("Maximum number = %d", max);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter the values of a ,b ,c 2 5 1
Maximum number = 5
PS C:\Users\Lenovo\Desktop\C>
```

19. PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDITIONAL OPERATOR.

Program_19.c > main()

```c
#include <stdio.h>
int main()
{
    char ch;

    printf("Enter tha character : ");
    scanf("%c", &ch);

    (ch>='a' && ch<='z') ? printf("Character is small letter") : printf("Character is not a small letter");

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Program_19.c -o Program_19 } ; if ($?) { .\Progr
Enter tha character : g
Character is small letter
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Program_19.c -o Program_19 } ; if ($?) { .\Progr
Enter tha character : S
Character is not a small letter
PS C:\Users\Lenovo\Desktop\C>
```

20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH CASE.



```
16      switch(ch)
17      {
18          case '+' :
19          printf("sum is = %.2f", a+b);
20          break;
21          case '-' :
22          printf("difference is = %.2f", a-b);
23          break;
24          case '*' :
25          printf("multiplication is = %.2f", a*b);
26          break;
27          case '/' :
28          printf("division is = %.2f", a/b);
29          break;
30      }
31
32      return 0;
33  }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\"
Enter a : 10
Enter a operator (+, -, *, /) : /
Enter b : 2
division is = 5.00
PS C:\Users\Lenovo\Desktop\C>
```

21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR) AND THIS PERFORM USING SWITCH CASE (CASE IN CHAR FORM).

Program_21.c > main()

```c
#include <stdio.h>
int main()
{
    float a, b, result;
    char ch;

    printf("Enter the first number");
    scanf("%f", &a);

    printf("Select operator(+,-,*,/) : ");
    scanf(" %c", &ch);

    printf("Enter the second number");
    scanf("%f", &b);

    switch(ch)
    {
        case '+':
        result = a+b;
        printf("result = %f", result);
        break;
```

```c
        case '-':
        result = a-b;
        printf("result = %f", result);
        break;

        case '*':
        result = a*b;
        printf("result = %f", result);
        break;

        case '/':
        result = a/b;
        printf("result = %f", result);
        break;
    }

    return 0;
}
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ;
Enter the first number 4
Select operator(+,-,*,/) : *
Enter the second number 5
result = 20.000000
PS C:\Users\Lenovo\Desktop\C>
```

22. PROGRAM TO CALCULATE FACTORIAL OF A NO.

Program_22.c > main()

```c
#include <stdio.h>
int main()
{
    int number, i=1;
    long int factorial;
    factorial = 1;

    printf("Enter a number : ");
    scanf("%d", &number);

    for(i = 1; i <= number; i++)
    {
        factorial = factorial * i;
    }

    printf("Factorial of %d = %ld", number, factorial);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter a number : 4
Factorial of 4 = 24
PS C:\Users\Lenovo\Desktop\C>
```

23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE (<MATH.H>).

Program_23.c > main()

```c
#include <stdio.h>
int main()
{
    int power, number, i;
    long int result;
    printf("Enter number : ");
    scanf("%d", &number);
    printf("Enter power : ");
    scanf("%d", &power);
    result = 1;
    i=1;
    while(i<=power)
    {
        result = result * number;
        i=i+1;
    }
    printf("The result is : %.2ld", result);
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ;
Enter number : 5
Enter power : 3
The result is : 125
PS C:\Users\Lenovo\Desktop\C>
```